

2. **Personal water treatment** – It is really necessary to have new technologies to treat water at its point of use. People will need to be equipped with devices that enable them to purify water at the tap, at the well, or in their residences. The idea is to dismantle the current model of centralized water-treatment plants and to replace it with small, strategically placed treatment systems that meet the water needs of population clusters. Such satellite treatment systems would be particularly useful in developing countries like India, where big plants are still rare and there is a tremendous need for cost-effective ways to bring clean drinking water to the masses. These satellite treatment systems could be customized to remove the specific contaminants found in a local water source. Nanosorbents, nanocatalysts, smart membranes, nanosensors and other kinds of nanotechnology could serve as the basis for new, small scale water treatment systems. The goal of personal water treatment might actually prove easier to reach than the goal of integrating nanotechnology into existing centralized water treatment plants operated by public utilities.
3. **Emerging pollutants.** Unfortunately, new kinds of pollutants are being continually discovered in water resources, even while we still deal with old problems such as lead and pesticides. Waterways now contain trace quantities of personal-care products such as sunscreens, medicines of all kinds and flame retardants and plastic residues that slough off consumer products. Some of these materials have been shown to have deleterious effects on fish and other wildlife, and they might also cause subtle health effects in people. Nano-enhanced water filtration could be developed to target these new contaminants. Advanced methods of water purification using nanotechnology might prove to be the only viable strategy for keeping a wide variety of nano-materials off the growing list of emerging pollutants.

Computers

The history of computers and computer technology thus far has been a long and a fascinating one, stretching back more than half a century to the first primitive computing machines. In the past twenty years, there has been a dramatic increase in the processing speed of computers, network capacity and the speed of the internet. These advances have paved the way for the revolution of fields such as quantum physics, artificial intelligence and nanotechnology. These advances will have a profound effect on the way we live and work, the virtual reality we see in movies like the Matrix, may actually come true in the next decade or so.

In the past twenty years, there has been a dramatic increase in the processing speed of computers, network capacity and the speed of the internet. These